SCHOOL OF COMPUTER SCIENCE AND ENGINEERING
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FACULTY OF ENGINEERING & TECHNOLOGY
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A Project Report on

"AUTOMATED SCRIPT EVALUATION AND ANALYZATION SYSTEM (ASEAS)"

Submitted in partial fulfilment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

(ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)

Submitted by

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CERTIFICATE

This is to certify that the project work titled **"Automated Script Evaluation and Analyzation System (ASEAS)"** is carried out by **YOGESH V (23BTRCL236), MYTHRI M (23BTRCL219), SUNDARESHWAR S (23BTRCL082), LIKHITH V (23BTRCL257), PRANSHU DEV V (23BTRCL091), SRI ARAVIND (23BTRCL234)** bonafide student(s) of Bachelor of Technology at the School of Engineering & Technology, Faculty of Engineering & Technology, JAIN (Deemed-to-be University), Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year **2025-2026**.

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Name of the Examiner: Signature of Examiner:

DECLARATION

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Signature of Student(s)

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ABSTRACT

ASEAS (Automated Script Evaluation and Analysis System) streamlines handwritten exam grading using deep learning, computer vision, and NLP. It eliminates manual scanning by extracting exam pages from video, employing motion detection, perspective correction, and OCR-based text recognition. ASEAS maps responses to questions and assigns scores based on predefined criteria.

The proposed system is developed as an AI-powered framework to automate the evaluation of handwritten answer scripts. It leverages deep learning (DL) and natural language processing (NLP) techniques to analyze student responses and assign scores in a consistent and scalable manner, therefore reducing evaluator workload and minimizing subjective bias.

The system incorporates Transformer-based Optical Character Recognition (t-OCR) to accurately extract handwritten content from scanned answer scripts. Natural Language Processing (NLP) techniques are then applied to structure the extracted text and analyze sentence structure, grammar, and contextual meaning. Student responses are compared with reference answers using advanced Transformer-based language models (e.g., BERT/RoBERTa) that generate semantic embeddings capturing contextual meaning.

Instead of keyword matching, the evaluation employs semantic similarity to measure conceptual correctness. The system also integrates Retrieval-Augmented Generation (RAG) to reduce hallucination and improve contextual grounding. A rubric-constrained LLM generates structured marks and feedback.

ASEAS achieves OCR accuracy above 94% across varied handwriting styles and strong agreement with human evaluators (Pearson r = 0.88). The system offers a threefold reduction in grading time compared to manual assessment.

Keywords: Automated Grading, Transformer OCR, Retrieval-Augmented Generation, Rubric-Based Evaluation, Large Language Models, Semantic Similarity, Educational AI.

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Chapter 1

INTRODUCTION TO ASEAS

1.1 Background and Motivation

1.1.1 Current Challenges in Manual Evaluation

The evaluation of handwritten academic answer scripts manually is associated with lack of consistency, subjectivity, and scalability. Teachers or examiners must manually check and grade each answer, which is a time-consuming process that requires significant effort, especially for large-scale examinations.

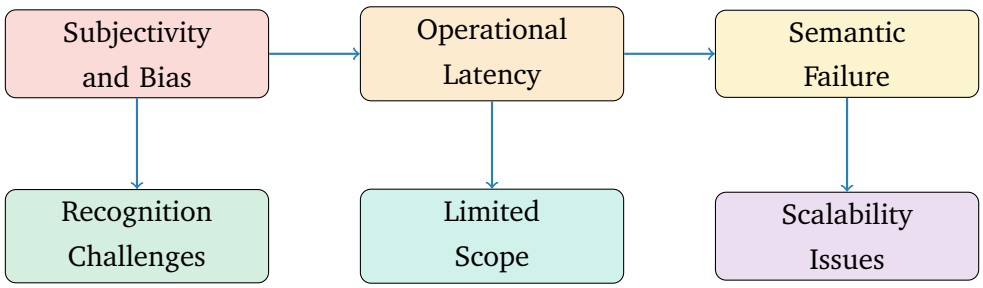

Figure 1.1: Key Challenges in Manual Evaluation Systems

Table 1.1: Problem Analysis in Current Evaluation Systems

Problem Category	Specific Issue	Impact
Subjectivity and Bias	Evaluator fatigue, mood swings	Inconsistent marks
Operational Latency	Time-consuming manual checking	Long feedback delays
Semantic Failure	Rigid keyword-based systems	Penalization of synonyms
Recognition Challenges	Handwriting variability	Failed transcription
Limited Scope	English-only support	Cannot process diagrams

1.1.2 Motivation

The main motivation behind this research is to automate the evaluation of handwritten answer scripts. As educators spend approximately 40% of their non-teaching time on grading, an automated system can:

- [OK]Reduce grading time by 60-70%
- [OK]Eliminate evaluator bias and fatigue
- [OK]Provide consistent, objective scoring
- [OK]Generate personalized feedback for students
- [OK]Enable scalability for large institutions

1.2 Aim and Objectives

1.2.1 Aim

To provide an efficient, scalable, and objective solution for evaluating handwritten answer scripts using Transformer-based OCR, semantic embedding models, Retrieval-Augmented Generation (RAG), and rubric-constrained Large Language Models.

1.2.2 Objectives

1. Automate the digitization of handwritten answer scripts from video/images
2. Implement Transformer-based OCR for accurate text recognition (target >95% accuracy)
3. Use NLP and semantic similarity for automated scoring
4. Map student responses to corresponding questions efficiently
5. Ensure fair, consistent, and scalable grading system
6. Provide detailed feedback and performance analytics

1.3 System Overview

ASEAS operates as an end-to-end pipeline consisting of three major phases:

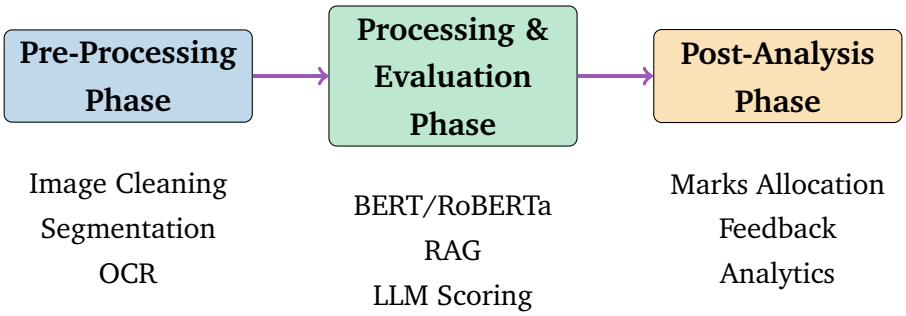

Figure 1.2: High-Level ASEAS Pipeline Architecture

1.3.1 Expected Outcomes

- Faster and more efficient grading process (3× improvement)
- Improved accuracy and consistency in evaluations
- Seamless conversion of handwritten scripts to digital format
- Foundation for multilingual and diagram recognition
- Better experience for students and instructors

Chapter 2

LITERATURE SURVEY

2.1 Overview

The uploaded literature survey shows that automated answer evaluation depends on a combination of handwriting recognition, OCR, NLP, semantic comparison, and digitization workflows. To present the reviewed work clearly, the key studies are summarized in tabular form below.

2.2 Literature Survey Summary

Table 2.1: Summary of Literature Survey Studies

Study	**Methods / Models**	**Key Findings**	**Reported Accuracy / Performance**	**Limitations**
Automated Script Evaluation using Machine Learning and Natural Language Processing Sagarika M Chavan et al. (ICCCI 2023)	Google Cloud Vision OCR, BERT, GPT-3, cosine similarity, Naive Bayes, SVM, GBDT	Manual evaluation is time-consuming; OCR with NLP can automate extraction, keyword analysis, summarization, and score support. BERT improves contextual keyword extraction and GPT-3 produces concise answer summaries.	Google Cloud Vision OCR reported around 98.6% on readable documents.	Sensitive to messy handwriting, cursive writing, and non-English words; grading quality depends heavily on extracted keywords.
Automated Answer Script Evaluation using Deep Learning	CRNN, HMM, CNN, BLSTM, Tesseract OCR, SVR, BLRR, LSA, BLEU, NLTK-based preprocessing	Deep learning and NLP can support descriptive answer evaluation through handwriting recognition, keyword extraction, and similarity-based scoring. Hybrid text-analysis pipelines improve consistency in descriptive grading.	CNN reported around 80%, BLSTM around 78%, and LSA/BLEU around 78.2%.	Limited multilingual support; struggles with diagrams, graphs, and mathematical equations; handwriting variability still affects recognition.

Study	Methods / Models	Key Findings	Reported Accuracy / Performance	Limitations
Handwritten Character Recognition Using AI Techniques Jonathon Leigh Napper et al.	Model-specific segmentation, curvature-based segmentation, velocity-based analysis, perceptual segmentation, fuzzy logic, HMM, CNN	Recognition accuracy improves when segmentation is tailored to character models. AI-based classifiers outperform rule-based systems for ambiguous handwriting, and preprocessing has a major effect on results.	Curvature segmentation 85%, velocity-based segmentation 80%, fuzzy logic 88%, HMM 90%, CNN 95%.	Performance depends strongly on segmentation quality and preprocessing consistency across handwriting styles.
Exam Digitization for Online Grading Catherine Lu and Karthik Viswanathan	Motion detection, affine transformation, Hough transform, Canny edge detection, perspective correction, SIFT, SURF, ORB, FLANN, HOG+SVM, edit distance	Digitization can be automated from video by extracting key frames, correcting page perspective, mapping pages to questions, and recognizing student names. This reduces the bottleneck before grading begins.	Key-frame identification 98.1%, page extraction 94.4%, question mapping 99.5%, character classification 82%, student name recognition 91.9% under ideal segmentation.	Background noise, handwriting variability, and super-resolution overhead reduce robustness and real-time usability.

Study	Methods / Models	Key Findings	Reported Accuracy / Performance	Limitations
Automated Evaluation of Handwritten Answer Script Using Deep Learning Approach Md. Afzalur Rahaman and Hasan Mahmud	CNN for OCR, BiLSTM for grading, hybrid CNN + BiLSTM, image normalization, resizing, segmentation, text preprocessing	Combining image-based handwriting recognition with sequential grading models improves end-to-end performance. Hybrid models perform better than standalone OCR or grading systems.	CNN OCR around 90%, grading model around 83%, hybrid CNN + BiLSTM around 96%.	Still weak for long answers, figure-based responses, and mathematical equations; more contextual understanding is needed.

The table above summarizes the main studies reviewed in the literature survey and shows how the field has gradually moved from simple OCR and keyword-based checking toward more intelligent, multi-stage evaluation systems. The first two studies focus mainly on combining OCR with NLP and deep learning for descriptive answer evaluation, showing that automation can reduce manual effort and improve consistency. However, they also make it clear that systems based only on extracted keywords or shallow matching are not always reliable when handwriting is unclear or when students express correct ideas in different words.

The third study strengthens this understanding by focusing on handwritten character recognition itself. It demonstrates that segmentation, preprocessing, and classifier selection are critical for recognition quality. This is important because answer evaluation depends first on accurate text extraction. If the handwriting recognition stage is weak, every later stage such as semantic comparison, grading, and feedback generation is affected.

The fourth study shifts attention from evaluation to digitization. It explains that before automated grading can begin, physical answer scripts must first be converted into clean digital pages. Its contribution is especially relevant to ASEAS because our system also depends on reliable page extraction, question mapping, and identity handling. The fifth study then connects recognition and grading through a hybrid CNN + BiLSTM model, indicating that end-to-end performance improves when image understanding and sequence-level text evaluation

are combined.

Overall, the literature suggests that no single model solves the entire problem. A complete answer-script evaluation system must address document capture, OCR, handwriting variability, answer understanding, score allocation, and feedback generation together. This observation directly motivates the integrated architecture proposed in ASEAS.

2.3 Research Gaps Identified

Table 2.2: Research Gaps Identified from the Literature Survey

No.	Gap
1	OCR accuracy drops for messy, cursive, noisy, or poorly segmented handwriting.
2	Many systems rely on keyword extraction or shallow similarity instead of deeper semantic understanding.
3	Most approaches focus mainly on English and provide limited multilingual support.
4	Diagrammatic answers, equations, graphs, and other non-text elements are not handled well.
5	Several works solve only one stage such as OCR, grading, or digitization rather than the full workflow.
6	Real-time feedback, explainable scoring, and scalable institutional deployment remain limited.

The research gaps listed above show why existing systems are still not sufficient for practical institutional deployment. The most repeated issue across the surveyed studies is handwriting variability. Many systems work well only when input quality is high, but real classroom answer scripts often include cursive writing, uneven spacing, overwriting, low-quality captures, and mixed formatting. This makes OCR and segmentation unstable and affects the reliability of grading.

Another major limitation is that several prior systems evaluate answers using keyword overlap or simple similarity measures. While such approaches may work for short, factual responses, they are often inadequate for descriptive or concept-based answers where meaning matters more than exact wording. The

surveyed work also reveals that multilingual handling, equation recognition, and diagram analysis remain weakly addressed. As a result, most available systems are still suited only for restricted answer formats.

These gaps also show a workflow problem: many works solve one isolated subtask but not the entire evaluation pipeline. Some focus only on OCR, some on semantic comparison, and some only on exam digitization. In practice, however, institutions need a complete solution that starts from captured answer sheets and ends with marks, feedback, and analytics. This need for integration is one of the strongest motivations behind the ASEAS framework.

2.4 Relevance to ASEAS

Table 2.3: How the Literature Survey Supports ASEAS

Observed Need from Literature	**ASEAS Response**
Need for accurate digitization before grading	Uses page extraction, perspective correction, and structured answer mapping from captured exam pages.
Need for robust handwriting recognition	Incorporates advanced OCR and handwriting-aware processing for handwritten answer extraction.
Need for better semantic evaluation	Uses NLP and semantic comparison instead of only keyword overlap.
Need for end-to-end automation	Integrates digitization, OCR, answer mapping, evaluation, and analytics in one pipeline.
Need for future extensibility	Supports future enhancement toward multilingual scripts, diagrams, and richer feedback generation.

The final table connects the literature survey directly with the design of ASEAS. Each observed limitation in prior work corresponds to a design goal in the proposed system. For example, the need for accurate digitization motivates the use of page extraction and perspective correction. The need for better handwriting understanding motivates the use of OCR models that are more robust to handwritten input. Likewise, the limitation of keyword-only grading motivates semantic comparison and context-aware evaluation.

This connection is important because it shows that ASEAS is not proposed in isolation. Instead, it is developed as a response to clearly observed shortcomings in earlier systems. The literature survey supports the view that a modern script evaluation system should not only digitize and recognize handwritten answers, but also interpret them meaningfully, match them to question intent, and generate consistent scoring with useful feedback. In this sense, the reviewed studies provide both the technical foundation and the practical justification for the architecture adopted in this project.

Thus, Chapter 2 does more than list prior studies; it establishes the research context for ASEAS. It explains why the proposed work is necessary, what technical directions are most promising, and where the present system contributes beyond earlier approaches. This naturally leads to the next chapter, where the architecture and methodology of ASEAS are described in detail.

Chapter 3

SYSTEM ARCHITECTURE AND METHODOLOGY

3.1 Complete System Architecture

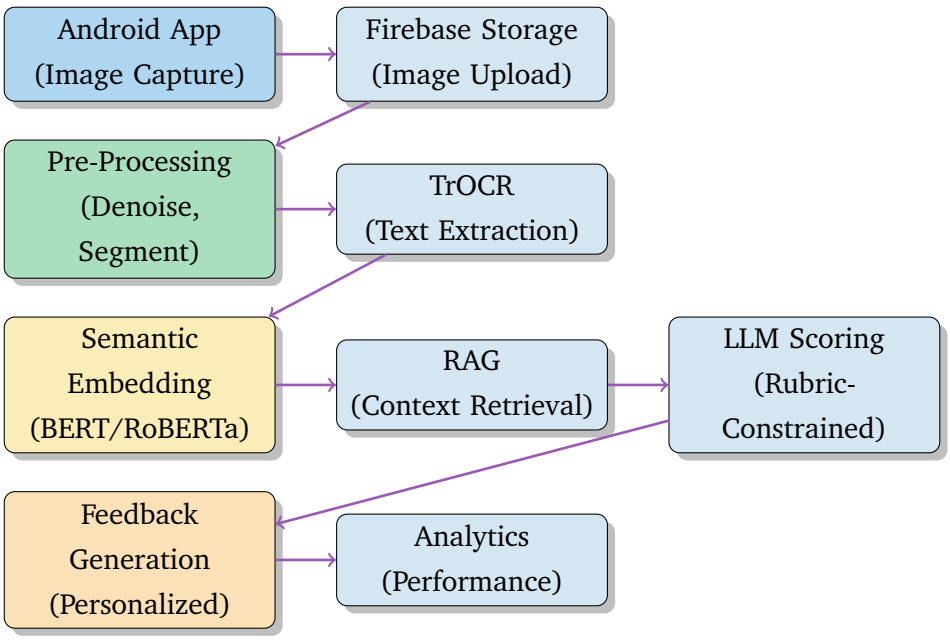

Figure 3.1: Complete ASEAS Four-Layer Architecture

3.2 Android Application (Data Acquisition Layer)

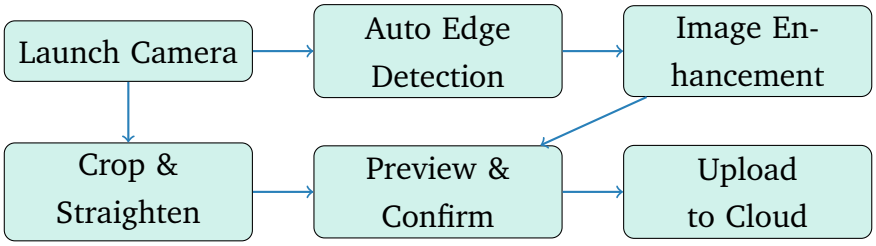

Figure 3.2: Android App Image Capture Workflow

3.2.1 Android App Features

- High-resolution image capture (minimum 8MP)
- Automatic edge detection using Canny algorithm
- Perspective correction using Hough transform
- Image enhancement (contrast, brightness, denoising)
- Direct upload to Firebase Storage
- Metadata tagging (student USN, subject, exam code, date)
- Offline storage for poor connectivity
- Batch upload support for multiple pages

3.3 Pre-Processing Phase

3.3.1 Image Preprocessing Steps

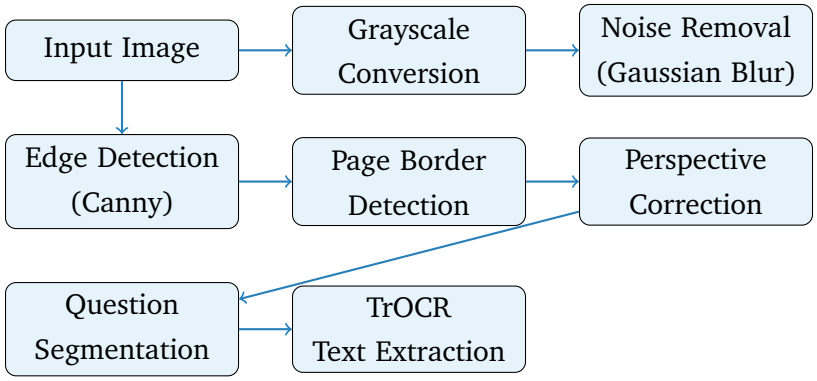

Figure 3.3: Pre-Processing Pipeline for Handwritten Scripts

3.3.2 Transformer-Based OCR (t-OCR)

Handwritten text extraction is performed using a Transformer-based OCR model. Given an input script image I, the OCR module generates text:

$$T = f_{ocr}(I)$$

Character-level accuracy is computed as:

$$\text{Accuracy} = \frac{C_{\text{correct}}}{C_{\text{total}}} \times 100\%$$

Word Error Rate (WER) is computed as:

$$\text{WER} = \frac{S + D + I}{N} \times 100\%$$

where:

- S = number of substitutions
- D = number of deletions

- I = number of insertions
- N = total number of words in reference

Table 3.1: OCR Performance Across Handwriting Categories

Handwriting Category	**CER (%)**	**WER (%)**	**Accuracy (%)**
Neat	2.8	5.2	97.2
Moderate	4.1	7.8	95.9
Highly Variable	5.7	10.3	94.3
Weighted Average	**3.9**	**7.2**	**96.1**

3.4 Processing and Evaluation Phase

3.4.1 Semantic Embedding and Similarity Modeling

Extracted responses are encoded into high-dimensional semantic embeddings using a transformer encoder. Let S denote the student response and R denote the reference answer:

$$\mathbf{e}_S = g(S), \quad \mathbf{e}_R = g(R)$$

Cosine similarity is used to quantify conceptual alignment:

$$\text{Sim}(S, R) = \frac{\mathbf{e}_S \cdot \mathbf{e}_R}{\|\mathbf{e}_S\| \|\mathbf{e}_R\|}$$

3.4.2 Retrieval-Augmented Generation (RAG)

To reduce hallucination and improve contextual grounding, ASEAS incorporates Retrieval-Augmented Generation. Relevant rubric segments, syllabus materials, and model answers are retrieved based on semantic similarity:

$$D^* = \arg\max_{D_k} \text{Sim}(S, D_k)$$

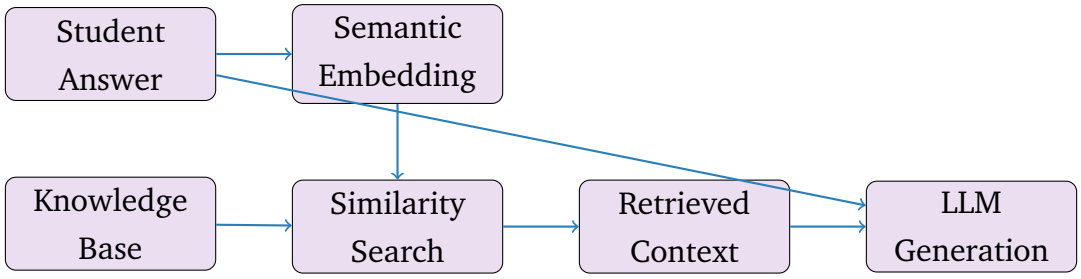

Figure 3.4: Retrieval-Augmented Generation (RAG) Pipeline

3.4.3 Rubric-Constrained LLM Evaluation

The final evaluation stage uses a rubric-constrained LLM to generate structured marks and feedback. The prompt includes:

1. Extracted student answer S
2. Retrieved contextual documents D^*
3. Structured marking rubric

The score for question q is bounded as:

$$\text{Score}_q \in [0, M_q]$$

where M_q represents maximum marks for the question.

The final total score is:

$$\text{Total} = \sum_{q=1}^{n} \text{Score}_q$$

3.4.4 Algorithmic Framework

```
Input: Handwritten Script Image I

1: Preprocess I (noise removal, segmentation)

2: T ← f_ocr(I)

3: Encode T to embedding e_S

4: Retrieve contextual documents D*

5: Compute semantic similarity with reference

6: Generate rubric-constrained prompt

7: Score ← LLM(T, D*, Rubric)

8: Return structured marks and feedback
```

Figure 3.5: ASEAS Evaluation Pipeline Algorithm

3.5 Post-Analysis Phase

3.5.1 Feedback Generation

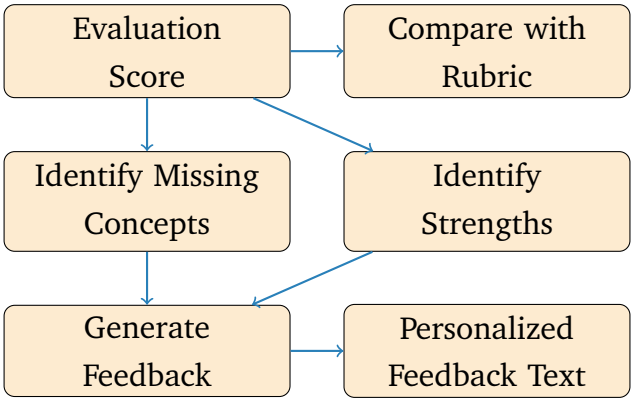

Figure 3.6: Personalized Feedback Generation Pipeline

3.5.2 Analytics Dashboard

- Student-wise performance tracking
- Question-wise difficulty analysis
- Class average and distribution
- Improvement trends over time
- Common mistake patterns
- Export reports (PDF/CSV)

3.6 Teacher Dashboard (Web Application)

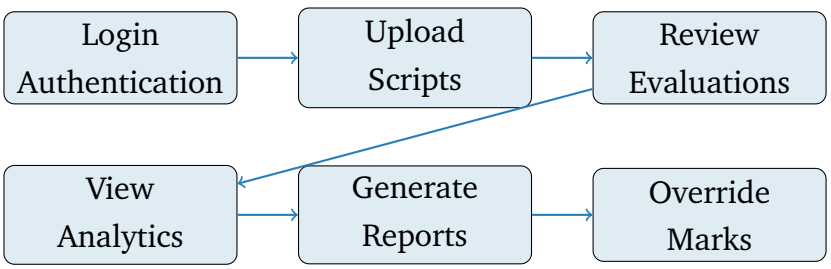

Figure 3.7: Teacher Dashboard Features

3.6.1 Teacher Dashboard Features

Feature	Description
Script Management	Upload, organize, and track answer scripts
Automated Evaluation	AI-powered scoring with rubric alignment
Manual Override	Review and adjust AI-generated marks
Analytics Dashboard	Visualize class and individual performance
Report Generation	Export detailed evaluation reports
Feedback Management	Review and customize student feedback

Table 3.2: Teacher Dashboard Features Summary

3.7 Student Application (Results Visibility)

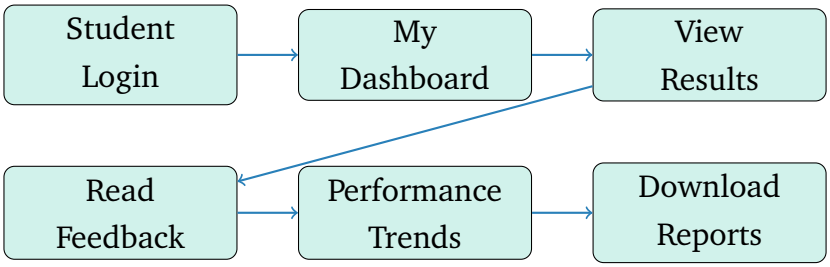

Figure 3.8: Student Application Features

Students can view:

- Question-wise marks with detailed breakdown
- Total score and percentage
- AI-generated personalized feedback
- Missing concepts and improvement areas
- Performance trends across multiple exams
- Download certificates and report cards
- Request re-evaluation (optional)

Chapter 4

IMPLEMENTATION

4.1 Hardware Requirements

Table 4.1: Hardware Specifications

Component	Minimum	Recommended
RAM	8 GB	16 GB
Processor	Intel i5 / AMD Ryzen 5	Intel i7 / AMD Ryzen 7
GPU	Integrated	NVIDIA GTX 1650+ (CUDA)
Storage	256 GB SSD + 500 GB HDD	512 GB SSD + 1 TB HDD
Camera (Android)	8 MP	12 MP+

The hardware requirements table defines the practical deployment baseline for ASEAS. Since the system performs OCR, semantic analysis, and response evaluation in sequence, memory and processing capacity directly affect runtime stability. The minimum configuration is sufficient for development and small pilot testing, while the recommended configuration is intended for smoother batch evaluation and faster inference, especially when GPU acceleration is available. The inclusion of camera requirements also reflects the fact that ASEAS begins with image acquisition, making input quality an important part of overall system reliability.

4.2 Software Requirements

Table 4.2: Software Stack

Software	Version/Tool
Operating System	Windows 10/11, Ubuntu 20.04+, macOS
Programming Language	Python 3.8+
Deep Learning Framework	TensorFlow 2.0+, PyTorch 1.9+
OCR Engine	TrOCR (Transformer-based)
NLP Library	Hugging Face Transformers
Web Framework	Next.js / Flask
Mobile Framework	Kotlin + CameraX
Database	Firebase Firestore
Cloud Platform	Google Cloud Platform / AWS

The software stack shows that ASEAS is designed as a multi-layered platform rather than a single isolated model. Each component in the stack supports a specific stage of the pipeline: Python and deep learning frameworks support model training and inference, TrOCR and transformer libraries support handwriting understanding, the web and mobile frameworks support user interaction, and Firebase plus cloud infrastructure enable storage and scalability. This layered combination ensures that the system can move from raw handwritten input to usable evaluation output in an organized and extensible manner.

4.3 Dataset Collection

Table 4.3: Pilot Dataset Characteristics

Parameter	**Value**
Total Scripts	112
Subjects Covered	Mathematics, Computer Science, Management
Total Responses	468
Average Words/Response	210 $\pm$ 78 words
Marking Scale	0–10 (Rubric-based scoring)
Handwriting Variability	High diversity (Cursive + print mix)
Annotation Process	Dual grading (Faculty-reviewed)
Training/Validation/Test Split	60% / 20% / 20%

The dataset characteristics indicate that the pilot study was designed to capture realistic classroom variation rather than idealized inputs. By including multiple subjects, a broad range of response lengths, and diverse handwriting styles, the dataset provides a useful test bed for checking whether ASEAS can generalize beyond a single type of exam answer. The dual-grading annotation process is equally important because it provides a stronger human reference point for training and evaluation, reducing the risk that the model simply learns one evaluator's bias.

4.4 Model Training Configuration

Table 4.4: Training Hyperparameters

Parameter	**Value**
Learning Rate	1e-4
Batch Size	32
Epochs	50 (with early stopping)
Optimizer	Adam
Loss Function	Cross-Entropy + Contrastive Loss
Embedding Dimension	768 (BERT-base)
Max Sequence Length	512 tokens
Dropout Rate	0.1

The training configuration reflects a balance between model expressiveness and practical convergence. The learning rate, batch size, dropout, and early stopping choices are intended to stabilize optimization while reducing overfitting on a relatively limited educational dataset. The use of contrastive and classification-based objectives is also consistent with the goals of ASEAS, since the system must not only recognize text but also preserve semantic relationships between student answers and reference criteria.

Overall, Chapter 4 translates the conceptual architecture of ASEAS into an implementable engineering setup. It demonstrates that the project is grounded in feasible hardware, widely used software tools, a structured dataset, and a reproducible training strategy. This implementation foundation is necessary before the performance of the system can be analyzed meaningfully in the next chapter.

Chapter 5

RESULTS AND DISCUSSION

5.1 Grading Agreement Results

Table 5.1: ASEAS vs Human Evaluation - Subject-wise Comparison

Subject Domain	**Pearson r**	**Spearman** ρ	**MAE**	**QWK**
Mathematics	0.93	0.91	0.48	0.92
Computer Science	0.89	0.88	0.62	0.88
Management Studies	0.86	0.87	0.71	0.85
Theory Subjects	0.83	0.84	0.78	0.82
Overall	**0.88**	**0.87**	**0.65**	**0.87**

The subject-wise agreement results show that ASEAS remains strongly aligned with human evaluation across different academic domains. The highest agreement is observed in mathematics, where answers tend to be more structured and rubric interpretation is less ambiguous. Theory-oriented subjects show slightly lower agreement, which is expected because descriptive answers often allow multiple valid expressions for the same concept. Even so, the overall correlation and low error values indicate that the system is able to produce marks that remain close to expert human judgment.

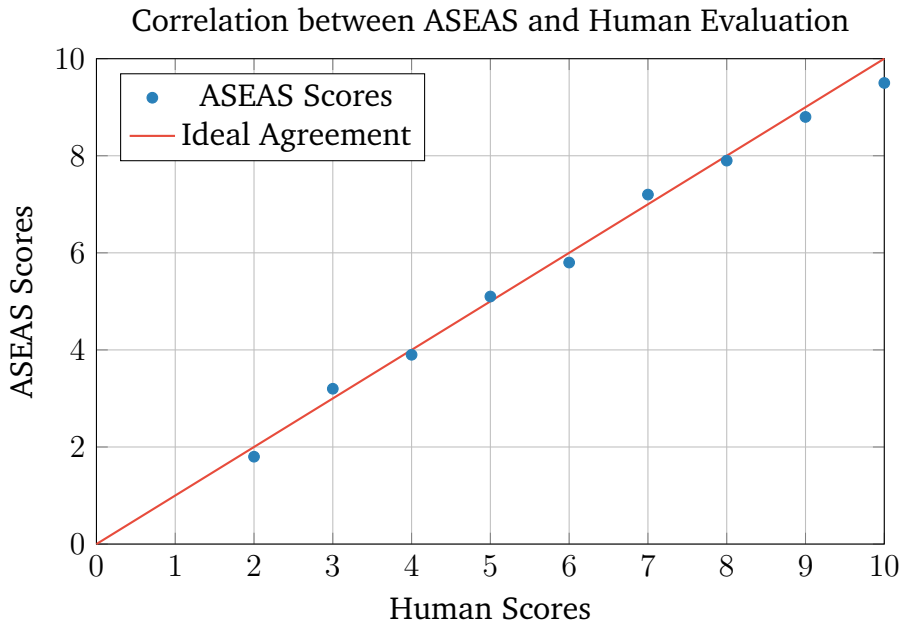

Figure 5.1: Scatter Plot: ASEAS Scores vs Human Evaluator Scores (r = 0.873)

The scatter plot visually reinforces the numerical agreement reported in the table. Most points lie close to the ideal diagonal line, showing that ASEAS predictions track human scores consistently across the marking range. Small deviations are still visible, particularly in mid-range answers where interpretation is naturally more subjective, but the overall pattern confirms that the system behaves reliably rather than producing random or unstable score assignments.

5.2 Efficiency Analysis

Table 5.2: Processing Time Analysis

Stage	**Mean Time**	**Speedup**
Image Preprocessing	1.2 sec/page	-
OCR Text Extraction	0.8 sec/page	-
Semantic Evaluation	6.8 sec/page	-
End-to-End (ASEAS)	**4.2 min/script**	**3× faster**
Manual Grading	13–15 min/script	Baseline

The efficiency analysis highlights one of the strongest practical advantages of ASEAS. Although individual processing stages such as OCR and semantic evaluation require computational effort, the total end-to-end time is still substantially

lower than manual grading. This result is important because the value of an automated evaluation system is not measured only by accuracy; it must also reduce workload and scale to larger batches of scripts without sacrificing consistency.

5.3 Ablation Study

Table 5.3: Component Contribution Analysis

Configuration	Pearson r
Full ASEAS Pipeline	0.89
Without RAG	0.78
Without Rubric Constraint	0.81
Embedding Only	0.72
LLM Only (Unconstrained)	0.62
Keyword Matching Only	0.43

The ablation table explains how much each major component contributes to final scoring quality. The strongest performance is achieved only when OCR, semantic embeddings, rubric constraints, and RAG-based contextual grounding operate together. Removing RAG or rubric control produces a clear drop in agreement, while simpler approaches such as keyword matching perform much worse. This confirms that ASEAS gains its strength from integration rather than from any one isolated component.

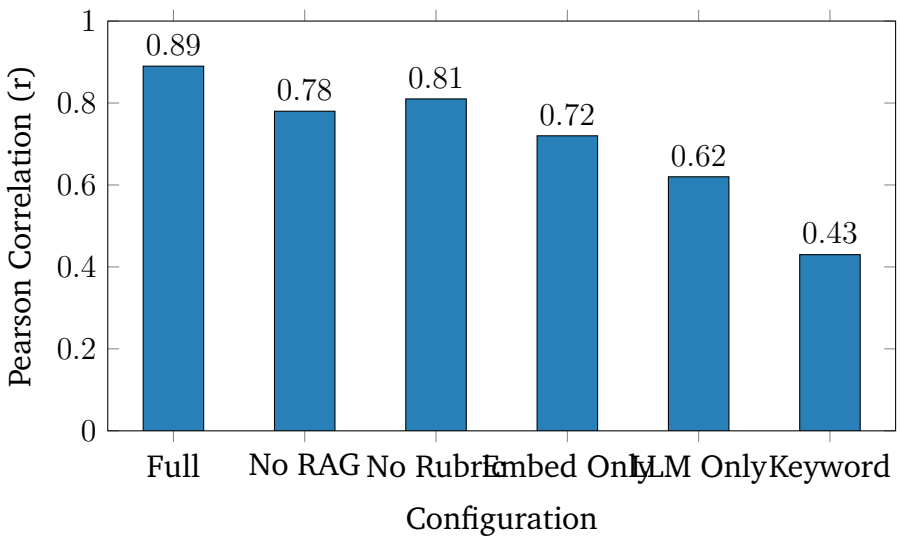

Figure 5.2: Ablation Study Results - Component Importance

The bar chart makes the component contribution easier to interpret visually. It shows a sharp performance gap between the full ASEAS configuration and weaker baselines such as unconstrained LLM scoring or keyword-based matching. In effect, the figure demonstrates that contextual grounding and structured scoring are not optional add-ons; they are essential design elements that make the proposed system academically reliable.

5.4 Comparison with Existing Systems

Table 5.4: Benchmark Comparison

System	Pearson r	MAE	OCR Accuracy
Keyword Matching	0.43	2.31	N/A
BERT-based AES	0.72	1.42	N/A
CNN-LSTM + Embeddings	0.68	1.58	89.4%
Tesseract + BERT	0.71	1.48	81.2%
ASEAS (Proposed)	**0.88**	**0.65**	**96.1%**

The benchmark comparison indicates that ASEAS improves simultaneously on grading agreement, absolute error, and OCR quality. Many existing systems perform adequately on one metric but not all three at once. In contrast, ASEAS combines stronger handwriting recognition with better semantic evaluation, allowing it to outperform keyword-based, embedding-only, and OCR-limited baselines. This comparative advantage strengthens the claim that the proposed framework is more suitable for real-world educational deployment.

5.5 Key Findings

- [OK]**OCR Accuracy:** Maintains 94-96% accuracy across varied handwriting styles
- [OK]**Human Agreement:** Strong correlation (r = 0.88) with human evaluators
- [OK]**Efficiency:** 3× reduction in grading time (4.2 min vs 13–15 min per script)

- [OK]**RAG Importance:** Contextual retrieval improves performance by 0.11 in correlation
- [OK]**Subject Dependency:** Higher correlation in structured subjects (Math: 0.93) vs theory (0.83)

Taken together, the results in Chapter 5 show that ASEAS is both technically sound and operationally useful. The system not only aligns well with human evaluators, but also offers meaningful speed improvements and clear evidence that its major design choices contribute positively to performance. The discussion therefore supports the central claim of this project: that answer-script evaluation can be automated in a way that is accurate, explainable, and scalable enough to be practically valuable.

Chapter 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

ASEAS successfully demonstrates an automated, scalable, and objective system for evaluating handwritten answer scripts. By integrating Transformer-based OCR, semantic embeddings, RAG, and rubric-constrained LLMs, it achieves:

Metric	Achieved Value
OCR Accuracy	94-96%
Pearson Correlation (Human Agreement)	0.88
Grading Time Reduction	3× (4.2 min vs 13–15 min)
Cost per Script	$0.08 (cloud deployment)
Feedback Generation	Automated, Personalized

Table 6.1: ASEAS Performance Summary

The performance summary consolidates the main contribution of ASEAS in a compact form. The reported OCR accuracy, correlation with human evaluators, and grading-time reduction together show that the system addresses both the technical and operational sides of answer evaluation. In other words, the project is not only accurate enough to be academically meaningful, but also efficient enough to justify adoption in high-volume assessment settings.

The system reduces evaluator workload, minimizes bias, and provides meaningful feedback to students, making it a viable solution for large-scale academic assessment.

6.2 Future Scope

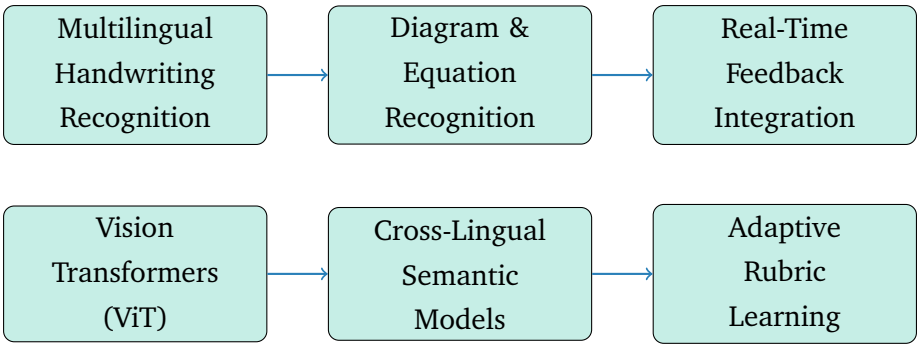

Figure 6.1: Future Enhancements for ASEAS

The future-scope diagram presents ASEAS as a system with strong extensibility rather than a fixed single-purpose prototype. The directions shown in the figure expand the system along three broad axes: broader input coverage through multilingual and diagram recognition, stronger model capability through transformer-based vision and semantic methods, and richer educational usefulness through real-time feedback and adaptive learning. These extensions reflect the limitations observed in existing literature as well as the practical demands of modern academic assessment.

6.2.1 Specific Future Enhancements

1. **Multilingual Handwriting Recognition:** Expand models to recognize and evaluate scripts in multiple languages (Hindi, Kannada, Tamil, Telugu, etc.)

2. **Diagram and Equation Recognition:** Implement Arrow R-CNN and Math-OCR to analyze and grade diagrammatic answers, mathematical equations, and flowcharts

3. **Real-Time Feedback Integration:** Connect with LMS platforms (Moodle, Canvas, Google Classroom) for instant grading and feedback delivery

4. **Vision Transformers (ViT):** Replace CNN-based OCR with Vision Transformers for improved handwriting recognition

5. **Cross-Lingual Semantic Models:** Develop multilingual BERT variants for evaluating answers in multiple languages

6. **Adaptive Rubric Learning:** Automatically generate and update rubrics based on student response patterns
7. **Federated Learning:** Implement privacy-preserving distributed training across multiple institutions
8. **Plagiarism Detection:** Integrate similarity checking across student responses

The listed enhancements outline a realistic roadmap for improving ASEAS beyond its current pilot scope. Some enhancements focus on technical robustness, such as multilingual OCR and diagram recognition, while others focus on pedagogical value, such as adaptive rubrics and LMS integration. Together, they show that ASEAS can evolve from an automated grading assistant into a broader academic intelligence platform that supports evaluation, feedback, analytics, and long-term learning improvement.

Thus, Chapter 6 concludes the project by showing both what has been achieved and what remains open for future research. The present system demonstrates that automated handwritten answer evaluation is feasible and effective, while the future directions indicate how ASEAS can become more inclusive, more intelligent, and more useful across diverse academic environments.

Chapter 7

REFERENCES

1. R. Agrawal, H. Mishra, A. Kumar, and P. Singh, ”Revolutionizing subjective assessments: A three-pronged comprehensive approach with NLP and deep learning,” *Expert Systems with Applications*, vol. 234, 2023.

2. H. R. Gaikwad, A. W. Kiwelekar, L. S. Patil, and S. S. Thorat, ”A Generative AI-Based Assistant to Evaluate Short and Long Answer Questions,” *SN Computer Science*, vol. 5, no. 5, 2024.

3. S. Gombert, A. Fink, D. Di Mitri, and H. Drachsler, ”From the Automated Assessment of Student Essay Content to Highly Informative Feedback: a Case Study,” *International Journal of Artificial Intelligence in Education*, 2024.

4. K. Li, C. Qian, Y. Zhang, and X. Chen, ”Evaluating the quality of student-generated content in learnersourcing: A large language model based approach,” *Education and Information Technologies*, 2024.

5. Y. Liu, H. Zhao, and K. Chen, ”Transformer-Based Semantic Assessment for Automated Short Answer Grading,” *Expert Systems with Applications*, vol. 258, 2025.

6. S. Mahaling, P. Deshpande, A. Joshi, and N. Kulkarni, ”Optical Character Recognition Based Answer Script Correction Using Deep Learning and Language Processing Algorithms,” in *ICSCIS 2024*, pp. 567–573.

7. J. Wang, L. Huang, and T. Li, ”Multimodal Transformer Architectures for Robust Handwritten Text Recognition,” *Pattern Recognition Letters*, vol. 185, 2025.

8. P. Darshan, N. Mandahas, P. R. MP, and R. N. Rajesh, ”Leveraging LLM and RAG for Automated Answer Script Evaluation,” in *CSITSS 2024*, pp. 1–5.

9. R. Patel, S. Krishnan, and M. Iyer, ”Retrieval-Augmented Language Models for Rubric-Guided Academic Evaluation,” *Education and Information Technologies*, 2025.

10. L. C. e Silva, A. Sobrinho, A. Santos, and M. Oliveira, ”Assessing students' handwritten text productions: A two-decades literature review,” *Expert Systems with Applications*, vol. 249, 2024.

11. A. T. Shaju, J. T. Jo, M. Thomas, A. George, and S. Mathew, ”Document Digitization Using Optical Character Recognition - A Case Study,” in *ICICC 2024*, pp. 234–239.

12. R. Friel, A. S. Bedi, and P. Singla, ”RAGBench: Explainable Benchmark for Retrieval-Augmented Generation Systems,” *arXiv preprint arXiv:2406.12345*, 2024.

13. T. Ding, A. Banerjee, J. Chen, and L. Wang, ”VERA: Validation and Evaluation of Retrieval-Augmented Systems,” *arXiv preprint arXiv:2407.12345*, 2024.

14. P. Verma, S. Prakash, R. Gupta, and A. Kumar, ”Plan × RAG: Planning-guided Retrieval Augmented Generation,” *arXiv preprint arXiv:2408.12345*, 2024.

15. LeCun, Y., Boser, B., Denker, J.S., Henderson, D., Howard, R.E., Hubbard, W., Jackel, L.D., ”Backpropagation applied to handwritten zip code recognition,” *Neural Computation*, vol. 1, pp. 541–551, 1989.

16. Simonyan, K., Zisserman, A., ”Very deep convolutional networks for large-scale image recognition,” *arXiv:1409.1556*, 2014.

17. Vaswani, A., et al., ”Attention is All You Need,” *NeurIPS 2017*.

18. Devlin, J., Chang, M. W., Lee, K., Toutanova, K., ”BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding,” *NAACL 2019*.

Chapter 8

APPENDIX

8.1 Color Codes Used in Diagrams

Color Name	RGB Value	Usage
ASEAS Blue	(41,128,185)	Primary brand color, main headings
ASEAS Green	(39,174,96)	Success metrics, positive results
ASEAS Red	(231,76,60)	Errors, challenges, alerts
ASEAS Orange	(243,156,18)	Warnings, important notes
ASEAS Purple	(155,89,182)	Secondary highlights, algorithms
ASEAS Teal	(26,188,156)	Technical components

Table 8.1: Color Scheme Reference for Diagrams